



# Three good links, one edge each

Second Edition, Short Run and Even Keel against each other and against today's link — the figures, then a recommendation

## What all three share

Ten designs are in the pack. These three offer the best overall balance; others keep a specific advantage, set out on page 2. On everything below the three are the same, and on each line better than or level with today — with one exception, leave, which is under the costs on page 2.

- ✓ All **9 of 9** December staffing rules, and all 3 flexible ones (today 4 of 9, 0 flexible)

✓ **7 full weekends off in 26**, never more than 5 weeks apart (today 4 in 20, up to 7 apart)

✓ Every closing shift **30 to 45 minutes shorter**; no shift over 9h 00m (today 9h 10m)
- ✓ **No fatigue warning**, fixed rota or with a cover week placed badly (today 4, up to 5)

✓ The same **35-hour** week and **218.6 days** a year, under the ceiling of 219 (today 219.0)

✓ **4 single rest days** in the rotation, as today; rest days otherwise in pairs or longer (22 of 26 breaks, today 13 of 17)

## Where they differ

The best of the three on each line is in bold green. Today is shown for scale, not ranked.

| MEASURE                                                     | TODAY              | SECOND EDITION            | SHORT RUN                 | EVEN KEEL          |
|-------------------------------------------------------------|--------------------|---------------------------|---------------------------|--------------------|
| Most days in a row, fixed rota                              | 7                  | 6                         | <b>5</b>                  | <b>5</b>           |
| Most days in a row, cover week placed badly                 | 9                  | 6                         | 6                         | 6                  |
| Heaviest week                                               | 43h 50m            | 42h 00m                   | 42h 30m                   | <b>41h 30m</b>     |
| Lightest week                                               | 25h 10m            | <b>25h 30m</b>            | 24h 30m                   | 24h 30m            |
| Shortest rest between shifts                                | 12h 30m            | 14h 20m                   | 14h 20m                   | <b>14h 35m</b>     |
| Weeks on one turn (one shift time Mon–Fri, no mixing)       | 7 of 16            | 18 of 21                  | 18 of 21                  | 18 of 21           |
| Weeks mixing earlies and lates                              | 7                  | <b>0</b>                  | <b>0</b>                  | 1                  |
| Longest block of earlies or lates (up to: with cover weeks) | 2 (up to 3)        | <b>5 (up to 7)</b>        | <b>5 (up to 7)</b>        | 5 (up to 12)       |
| Week-to-week change of start time                           | 4h 00m             | <b>1h 01m</b>             | 1h 13m                    | 1h 16m             |
| Finishing at 23:00 or later, each a year                    | 39                 | <b>42</b>                 | <b>42</b>                 | 44                 |
| Shift times to learn that are new                           | 0 of 18            | <b>6 of 15</b>            | <b>6 of 15</b>            | 7 of 15            |
| Fit to the trains — weekday · Sat · Sun (0 = exact)         | 52.0 · 65.9 · 71.7 | <b>29.1 · 12.7 · 29.0</b> | <b>29.1 · 12.7 · 29.0</b> | 29.4 · 14.5 · 29.0 |

## Each one's edge

|                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Second Edition</b><br/>SE-26-F1 · dea6417f</p> <p><b>The steadiest.</b> One shift time Monday to Friday in 18 of 21 weeks and the gentlest week-to-week change of the three, 1h 01m. Its cost is the one the other two have shed: 6 days in a row on the fixed rota.</p> | <p><b>Short Run</b><br/>SR-26-F1 · 618348d6</p> <p><b>The shortest stretches.</b> Second Edition's duty table on a different layout of rest days — the same times, the same 18 weeks on one turn — with no run over 5 days on the fixed rota. The price is one week at 42h 30m (Second Edition 42h 00m) and a slightly rougher step.</p> | <p><b>Even Keel</b><br/>EK-26-H1 · 8e9a1bcf</p> <p><b>The lightest heaviest week, and the most rest between shifts.</b> A different duty table: no week over 41h 30m, never less than 14h 35m between shifts (the others 14h 20m). It pays with 44 late finishes a year (the others 42), 7 new times, 1 mixed week, and a block of lates up to 12 weeks with its cover weeks.</p> |
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## Which one, by what matters most

| IF THE MAIN PRIORITY IS                                               | CHOOSE                                    |
|-----------------------------------------------------------------------|-------------------------------------------|
| Five-day runs on the fixed rota, and no week mixing earlies and lates | <b>Short Run</b>                          |
| The lightest heaviest week, and the most rest between shifts          | <b>Even Keel</b>                          |
| The steadiest start times from week to week                           | <b>Second Edition</b>                     |
| Four weeks off for 14 days' leave, as today                           | <b>Long Break</b> · outside the shortlist |

Every figure is computed from the rotas and checked independently; how each design was found is in the technical notes.

# Against today, and a recommendation

On most of what a colleague feels the three are the same — a figure in grey is shared by all three

## Against today's link

| MEASURE                             | TODAY       | SECOND EDITION | SHORT RUN   | EVEN KEEL   |
|-------------------------------------|-------------|----------------|-------------|-------------|
| Full weekends off, in the rotation  | 4 in 20     | 7 in 26        | 7 in 26     | 7 in 26     |
| Full weekends off, about a year     | 10          | 14             | 14          | 14          |
| Longest wait between full weekends  | 7 weeks     | 5 weeks        | 5 weeks     | 5 weeks     |
| Contracted days at work a year      | 219.0       | 218.6          | 218.6       | 218.6       |
| Saturdays a year, each              | 26          | 28             | 28          | 28          |
| Sunday overtime duties a year, each | 21          | 20             | 20          | 20          |
| Starting at 06:20, each a year      | 62          | 60             | 60          | 60          |
| Average shift, Monday to Saturday   | 8h 14m      | 8h 16m         | 8h 16m      | 8h 16m      |
| Longest shift                       | 9h 10m      | 9h 00m         | 9h 00m      | 9h 00m      |
| 14 days' leave buys, at best        | 30 days off | 28 days off    | 28 days off | 28 days off |
| 14 days' leave buys, at worst       | 19 days off | 20 days off    | 20 days off | 20 days off |
| Leave for four full weeks off       | 14 days     | 15 days        | 15 days     | 15 days     |

Per-year figures are the rotation's count scaled to 52 weeks; days a year is Monday to Saturday, a cover week counting four. Sundays stay overtime.

## Why these three and not the other seven

Ten designs are in the pack. These three offer the strongest overall balance under the priorities used here — days in a row, weekends, one shift time a week, late finishes. Some of the others keep a specific advantage: Fine Tune a step seven minutes gentler than Even Keel's, for two fewer weeks on one turn; Second Look fourteen distinct times rather than fifteen, for a heavier week; Long Break four weeks off for 14 days' leave and a 29-day best stretch, for a fifth single rest day, two fewer one-turn weeks and a 1h 44m step. The rest give something up for nothing the three lack.

## The honest costs, shared by all three

- **Late finishes.** 42 a year each at 23:00 or later (today 39), 44 on Even Keel. It is the December staffing: three to the close every day where today's weekdays have two, so no link meeting the rules can have fewer than 42.
- **Leave.** The best 14-day stretch is 28 days off, not today's 30; four full weeks off, Sunday to Saturday, takes 15 days' leave, not 14. The worst stretch is a day better. All three are worse than today here, on something people plan a year around; Long Break alone keeps the 14.
- **Saturdays.** About 2 more a year each (28, today 26) — fourteen on a Saturday, the flexible rule, shared across 26 lines (today ten across 20).
- **New times.** 6 of 15 shift times are new on Second Edition and Short Run (9 are worked today), 7 of 15 on Even Keel. Several of the new ones are shorter versions of today's lates: the weekday and Sunday closers start later, the Sunday office late finishes earlier.

## Recommendation

**Short Run.** It is Second Edition's duty table laid without the six-day stretch: 6 days in a row on Second Edition's fixed rota, never more than 5 on Short Run's (six for all three if a cover week falls badly). What it gives up is a half hour on the heaviest week (42h 30m against 42h 00m, both under today's 43h 50m), an hour off the lightest (24h 30m against 25h 30m) and 0h 12m on the week-to-week step. The recommendation gives most weight to avoiding a sixth day in a row, a cost on the fixed rota every cycle; a colleague who weighs steadier start times or a lighter heaviest week more highly may reasonably prefer Second Edition.

**But the choice is closer than the table above makes it look.** Weight every line and the three finish within about a point of a hundred, and which is first turns on whether the run is measured on the fixed rota or on a cover week placed badly. So choose by the edge that matters here:

- **Choose Short Run** if days in a row is what colleagues raise first.
- **Choose Even Keel** if the heaviest week and the shortest rest matter more than two late finishes a year and a seventh new time; the two late finishes buy a fourth person to the Saturday close. It is the only one on a different duty table — a genuine alternative, not a variant.
- **Choose Second Edition** if steady shift times week to week carry the day: the gentlest change of the three and a lighter heaviest week than Short Run, at the price of the six-day stretch.